



Society of Petroleum Engineers

SPE-227018-MS

Effect of Water with Dissolved CO₂ on Rock Properties: From Static Aging to Dynamic Flowing

Mohammed Fadhel Al-Hamad and Olivier Sindt, SLB, KSA; Shouxiang Mark Ma, Saudi Aramco

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227018-MS](https://doi.org/10.2118/227018-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Deep aquifers have been considered as good candidates for CO₂ sequestration due to their massive capacities. They have been evaluated extensively not only for storage capacity but also for integrity since CO₂ dissolved in formation-water forming weakly acidic carbonated water which can interact with formation rock matrix, especially carbonates. Many studies conducted in this regard were performed under static conditions, i.e., rock samples were exposed to carbonated water in a sealed vessel for a specific duration. This paper investigates the effects of carbonated water on rock properties under continuous dynamic flowing conditions and compares them with those obtained from a previous study under static ageing.

In this study, carbonate rock samples were prepared and loaded in a controlled core flooding system, where carbonated water was continuously injected. The changes in rock properties and mechanical integrity vs. time were monitored by means of nuclear magnetic resonance (NMR), resistivity, and ultrasonic measurements. Changes in rock morphology were also captured through high resolution surface micrographs. This process of injecting carbonated water and monitoring changes in rock properties lasted for a total of 6 months. The effluents were also collected during the experiment and analyzed geochemically, using a capillary electrophoresis system.

The experimental results showed that changes in rock properties with time of carbonated water injection was continuous, as indicated by NMR and resistivity measurements. From the NMR results, the rock pore surfaces and networks were likely altered due to exposure to carbonated water. The resistivity results revealed that the pore network is getting more complex with increased Archie porosity exponent. High resolution imaging and image analysis indicate that the rock surface morphology was also altered with observed partial dissolutions. The analyzed water effluents confirmed mineral dissolution with excess of calcium and magnesium cations. The ultrasonic measurements showed changes in rock stiffness, especially for calcite-rich carbonates. The above changes are more pronounced during continuous flowing when compared to those observed during static ageing.

Introduction

Carbon capture and subsurface storage (CCS) has been considered as a serious contender to reduce the CO₂ concentration in the atmosphere and therefore minimize global warming. Due to their much larger storage capacities, saline aquifers are good candidates for CO₂ sequestration (Zhang and Agarwal, 2012), though sequestration of CO₂ in oil and gas reservoirs, especially for CO₂ EOR as part of CO₂ subsurface utilization and storage, has a much longer history and studied more extensively (Holloway and van der Staaten, 1995; Hamza et al, 2021; Li et al, 2021; Karvounis and Blunt, 2021). When CO₂ is injected in subsurface formations, it is trapped in the following four mechanisms (Siqueira et al., 2017):

- Structural physical trapping: trapped by the overlying impermeable layer(s),
- Capillary residual trapping: trapped as immobile CO₂ in the pores systems of the rock by capillary pressure,
- Solubility trapping: trapped as dissolved species in the aqueous phases, and
- Mineral trapping: trapped as a solid mineral phase through mineralization, such as in carbonate formations.

Note that the solubility of CO₂ in brine is a factor which drives the efficiency of solubility trapping mechanism (Steel et al., 2016). Though it is affected by water salinity and solution gases (Wang and Ehlig-Economides, 2023), but once CO₂ is trapped as an aqueous component, it would be prevented from leaking and driving upwards to create a gas cap (Rosenbauer et al., 2002). But, until the stored CO₂ is mineralized, one major challenge in successful subsurface CO₂ storage is to maintain the stored CO₂ in the targeted formations permanently because the injected CO₂ will mix with formation water forming weakly acidic carbonated water which will change the partition of most dissolved species (Gaus, 2010). In carbonate rock reservoirs, this will lead to a disequilibrium of pore fluid chemistry and result in the aqueous phase being under-saturated with regards to carbonate minerals, leading to the dissolution of calcite, dolomite, and magnesite. With sufficient dissolution and precipitation, the original rock surfaces (Singer et al., 2023 and 2024) and pore network (Ma et al., 1993) that affect and regulate fluid distribution (Ma et al., 2006) and flow will change accordingly, affecting rock mechanical and petrophysical properties (Bemer and Lombard, 2010; Bacci et al., 2011; Ojala, 2011; Alam et al, 2014; Major et al, 2018; Kiseok et al, 2018; Wang et al, 2022; Al-Hamad et al., 2025a), i.e., carbonated water interacts with rock solids result in.

1. dissolution of minerals,
2. precipitation of minerals,
3. alteration of rock texture, pore structure, and pore surface properties,
4. changes in rock petrophysical and mechanical properties

Operationally, these reactions may have impacts on the processes of CO₂ injection and storage (Rötting et al., 2015), such as formation thinness and permeability, as well as the stress limit and fracturing pressure of the formations (Ambrose et al., 2008; Burke, 2011). In addition, they may also affect the integrity of both reservoir and caprock over large distances and for a long period of time (Gaus, 2010).

In our previous study (Al-Hamad et al., 2025a), the CO₂ carbonated water-rock interaction was investigated by dissolving CO₂ in brine and then ageing carbonate outcrop samples in the carbonated brine. The ageing mechanism was static; i.e., the core samples were kept with carbonated water for a given period of time in a sealed vessel (Singer et al., 2024). Despite ageing over extended periods of time (up to 6 months), results showed only limited degree of change in the measured rock properties. The rock mechanical integrity and stiffness were maintained, as illustrated by the minor reduction in shear modulus. In addition, it was also observed that changes seemed to happen quickly, during the first month of exposure of the rock sample to

CO₂ carbonated water. After that, the rock properties maintained for additional interaction. It is believed that since these experiments were run in a closed system, with a fixed amount of CO₂, the interaction between the carbonated brine and the rock minerals stopped once the reactive power of the CO₂ was consumed. This current study aims to verify that hypothesis and investigate the effect of dynamic continuous CO₂ flooding on rock properties and mechanical integrity, i.e., from the past static ageing (Singer et al., 2024; Al-Hamad et al., 2025a) to dynamic flowing reported in this study.

Experimental Section

Core Samples

Two outcrop carbonate rock samples were selected for the study: Indiana Limestone (IND) and Dolomite (DOL). The XRD mineralogy results showed that IND is mainly calcite with minor impurities (Al-Hamad et al., 2025b), and DOL contains approximately 87% dolomite, 11% quartz, and 2% gypsum. The initial properties of the samples are reported in Table 1, with helium porosity measured by pressure falloff technique using the CoreTest AP-608 machine.

Table 1—Petrophysical properties of the samples studied

Sample	L [cm]	D [cm]	Porosity [%]
IND	4.92	3.81	19.15
DOL	4.84	3.80	15.99

Fluids

A brine of the composition listed in Table 2 (total equivalent NaCl salinity of 50 kppm) was used in the study. The carbonated brine was prepared by dissolving CO₂ gas into the synthetic brine. The brine density was measured at 1.0386 g/cc and its viscosity was measured to be 1.0 cP, both at 25°C.

Table 2—Brine composition

Salt	NaCl	CaCl ₂	MgCl ₂	H ₂ O
g/l	41.75	9.05	3.20	946.0

Experimental Protocol

The rock samples underwent different petrophysical tests, including porosity permeability, acoustic velocity, resistivity and NMR, before and after being subjected to exposure of CO₂ carbonated brine. The porosity of the samples was initially measured at dry state using helium, after which the samples were saturated fully with the brine, followed by other tests (Fig. 1).

The samples were then loaded into a high-pressure high-temperature (HPHT) system. A net pressure of 1,500 psi and a temperature of 90°C was applied. After reaching equilibrium, the carbonated brine was injected, slowing to allow interaction between the fresh carbonated brine and rock matrix, through the core samples for different periods of duration; 15, 30, 60, 90, 120, 150 and 180 days, at a constant flowrate of 0.01 cc/min.

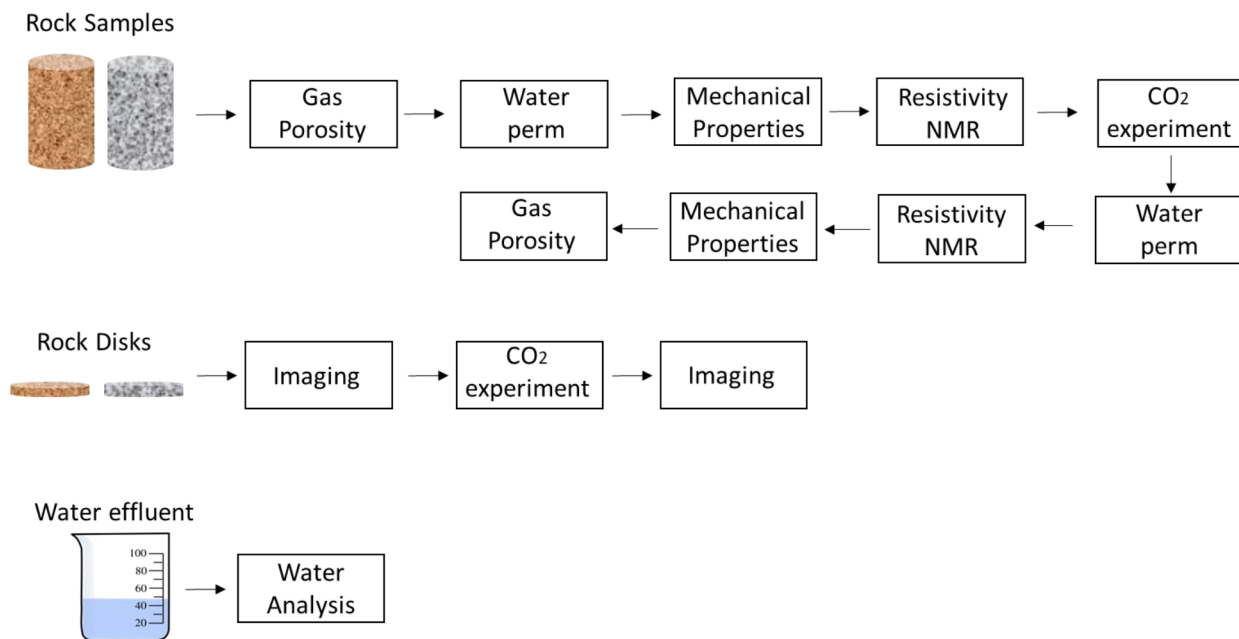


Figure 1—Experimental workflow

After each time period of carbonated brine injection, the samples were flooded by a native fresh brine (with no CO₂) for 5 pore volumes to displace all carbonated brine, and then permeability, acoustic velocity, resistivity and NMR were re-measured. Note that sample cleaning was only performed at the end of the 6 months period to measure the helium porosity.

To visually observe and study the effects of carbonated brine to rock samples, rock disk samples were prepared and treated with carbonated brine, under the same conditions as the core plugs, before being utilized for high resolution imaging trying to capture changes on the rock surfaces due to exposure to carbonated water. Effluent water samples were also collected during the experiment to analyze ionic concentration changes from rock dissolution. Details of the experimental workflow are shown in Figure 1.

Permeability Measurements

Brine permeability of core plugs was measured before and after CO₂ injection. Again, after each period of carbonated water flooding, the samples were flushed thoroughly with fresh non-CO₂ brine, under a net pressure of 1,500 psi and a temperature of 90°C, for 5 pore volumes. Then, brine permeability was measured with a flow rate of 0.25 cc/min while monitoring the pressure drop between the inlet and outlet. Permeability was calculated using Darcy's law,

$$K = \frac{q\mu L}{A\Delta P} \quad (1)$$

Where A is core plug cross-sectional area (cm²), K permeability (Darcy), L core length (cm), ΔP pressure drop between inlet and outlet (atm), q flow rate (cm³/sec), and μ brine viscosity (cp).

Resistivity Measurements

Similarly, core plug resistivity was also measured before and after exposure to carbonated water, while cores were fully saturated with fresh non-carbonated brine. The measurements were done using Fluke RLC resistivity meter, at 20 KHz using a 4-poles configuration (Ma et al., 2013). The Archie cementation factor m was derived from these measurements.

NMR Measurements

To test effects of carbonated water on rock surfaces and pore structure (Singer et al., 2024), NMR measurements were performed with an Oxford 2 MHz NMR spectrometer. T2 measurements were performed using Carr-Purcell-Meiboom-Gill (CPMG) with an inter echo time of 200 μ s. Pore sizes are partitioned with T2 cut-off values often used in the industry, i.e. $T2 \leq 10$ ms for microporosity and smaller pores, $10 < T2 < 100$ ms for medium pores and $T2 \geq 100$ ms for larger pores.

Acoustic Velocity Measurements

Core mechanical properties were measured and analyzed by means of ultrasonic velocity measurements (Syed et al., 2024). These were performed in a confining cylindrical cell. The compressional (P) and shear (S) waves were excited from one end of the sample and received at the other end by means of piezoelectric transducers. The measured P and S wave velocities (V_s and V_p) were used to compute the dynamic elastic properties, the shear and bulk modulus.

Effluent Water Chemical Analysis

Water samples of flooding effluent were collected and analyzed by using a CAPEL-205 capillary electrophoresis unit equipped with a UV/VIS detector supplied by LUMEX. The analysis was done using the cation program set in the ELFORUN software. The fresh carbonated brine sample was first analyzed before injection (and interaction) with rock samples. Collected effluents during the injections were analyzed, helping to identify if rock mineral dissolution has occurred, by looking at the concentration of ions captured via analyzing the collected water effluents.

To perform the water analysis, the instrument was conditioned and cleaned after each test, to ensure that the capillary is clean, free from contaminants, and ready for reproducible tests. Once the capillary was clean, the analysis was done using a cation program. The capillary electrophoresis accepts maximum concentration of 2,000 ppm, therefore all collected water samples were diluted by 25 times, using deionized water before analysis.

Results and Discussion

Permeability

Brine permeability of the core samples was measured before and after the various injection/ageing with carbonated water. The initial permeability measured for IND was 40.0 mD and 102.8 mD for DOL. In general, it was observed that contacting with carbonated water did alter the permeability of both rock types (Fig. 2). For the calcite IND sample, the major reduction in permeability was found to occur in the first 60 days of exposure to carbonated water, after which lower reduction in intensity was observed. A total reduction of 10.6 mD for an injection period of 6 months was measured, which could be attributed to the extraction of fine grains that clogged the pore throats. Similar observations on calcite-rich formations were also reported by others (Mohamed and Nasr-El-Din, 2013; Kim et al., 2018; Endar et al., 2022).

A completely different behavior was observed for the DOL sample that the permeability increased with injection of carbonated water. Like the calcite IND sample, the largest changes happened quickly at the beginning of the injection. It increased by 16.9 mD (a 16% increase) during the first 15 days of flooding, followed by an additional 6.6 mD increase for the rest of injection period. This phenomenon of increase in permeability after interactions with carbonated water was also observed by others (Tariq et al., 2018; Afkhami Karaei et al., 2020). Specifically, after carbonated water flooding dolomite rocks with permeability >100 mD, Holm (1959) found a threefold increase in permeability, attributed to dissolved dolomite minerals that created/enlarged pore throat sizes (Ma and Morrow, 1996).

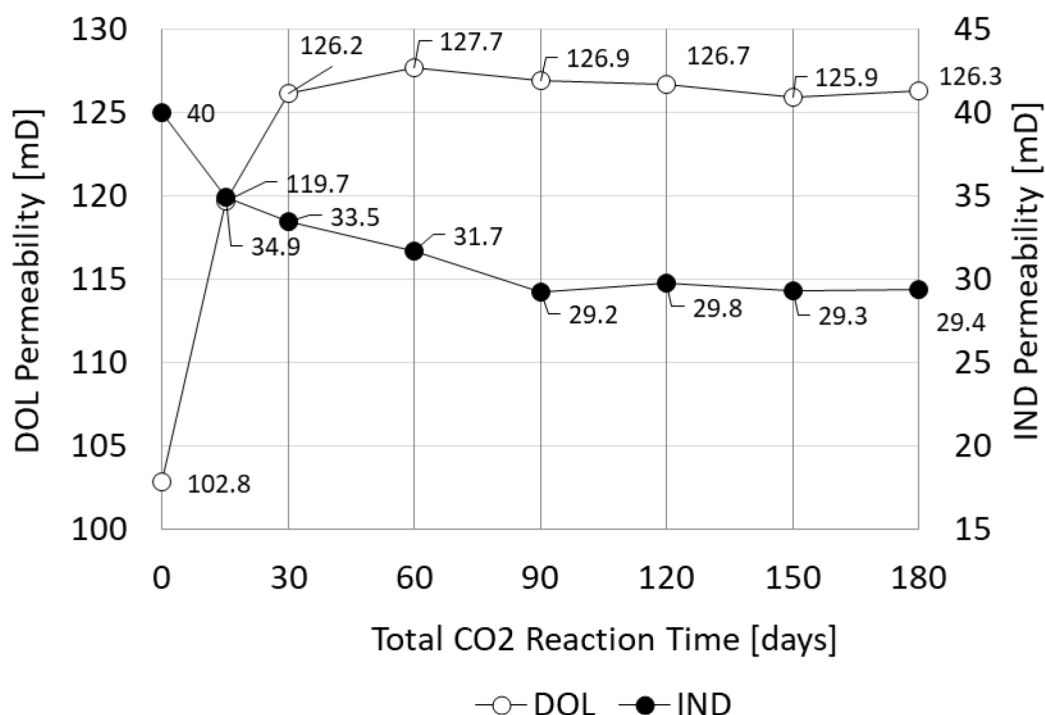


Figure 2—Effects of carbonated water flooding on rock permeability of dolomitic DOL and calcite IND samples

In this study, the increase in permeability may also be caused by the presence of gypsum and its solubility in carbonated water (Rukuni et al., 2015). Although the amount of gypsum in the DOL sample is small, 2% from XRD test, its dissolution could be sufficient to enhance permeability, depending on its distribution. From Figure 2, unlike the IND sample, clogging the pore throats thus reducing permeability was apparently not issue, likely due to its higher permeability, or larger connected pore throats, as compared to the IND sample (Lala and El-Sayed, 2017).

The dynamic induced permeability alteration of both IND and DOL samples was more significant than that of static ageing, based on the findings from our pervious study (Al-Hamad et al., 2025a). At static ageing, IND and DOL sample showed only 1.3mD and 0.48mD reduction in gas permeability.

Resistivity

Resistivity of non-carbonated brine (R_w) was first measured at 25°C and found to be 0.14 ohm.m. The resistivity of the core samples, under 100% brine saturation, R_o , was measured before and after carbonated water flooding. For the IND sample, $R_o=4.3$ ohm.m, and for DOL, it was 9.9 ohm.m. The lower resistivity for IND could be due to the higher porosity (19%, Table 1)) as compared to DOL (16%), though DOL has higher permeability (Fig. 2).

After each carbonated water flooding period, the sample resistivity was re-measured. Like permeability test (Fig. 2), a rapid change in resistivity was observed and the change was more pronounced in the early stages of carbonated water exposure (Fig. 3), i.e., for both samples, major changes were during the first 60 days of carbonated water flooding, after which the resistivity seems to plateau. A total resistivity increase of 60% and 41% was measured for IND and DOL samples, respectively. Unlike permeability data however, both samples show increases in resistivity, which would indicate decrease in permeability, like observed from the calcite IND sample, but not the dolomitic DOL sample. A more detailed study is needed to investigate this further.

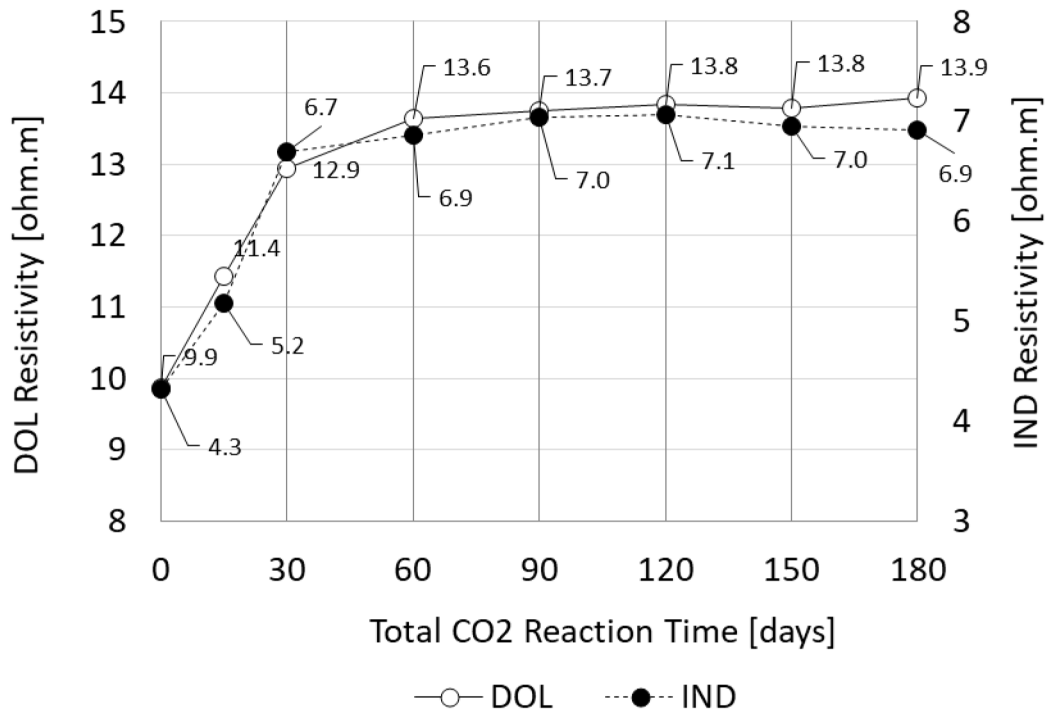


Figure 3—Effect of carbonate water flooding on rock resistivity

After the last step of the carbonated water flooding, the samples were cleaned, dried and helium porosity (ϕ) was re-measured. For the IND sample, the porosity was remeasured which reduced from the initial 19.15% (Table 1) to 18.85%, a drop of 0.3%. For the DOL sample, however, the remeasured porosity increased from its initial 15.99% (Table 1) to 16.55%, a 0.56% enhancement. The observed changes in porosity indicate that contacting with carbonated water did altered rock pore structure in terms of volume, and also the pore connectivity, as indicated by the permeability and resistivity measurements.

With measured resistivities and porosity, Archie cementation exponent (m) can be calculated,

$$\frac{R_o}{R_w} = \phi^{-m} \quad (2)$$

The calculated results showed that interaction between rock and carbonated water during the above tests increased m by about 10% and 13% respectively for the DOL and IND samples, implying more complex rock pore networks.

Comparing the resistivity data in both types of experiments; static (Al-Hamad et al., 2025a) vs. dynamic (this study), the changes in DOL formation were comparable, i.e., an increase of approx. 40%. For the IND formation, the dynamic continuous flow of carbonated brine had a more pronounced effect (60% increase in resistivity) than under static ageing (11% increase).

NMR

NMR T_2 of the rock samples was measured before and after each carbonated water flooding test. The T_2 distributions of the samples before ageing are shown in blue in Figures 4a and 4b. The initial IND sample shows a typical bimodal T_2 distribution in the range of 10 ms to 2200 ms, indicating mostly medium to large pores. The sample has a T_2 log-mean of around 200 ms. For the DOL sample, however, in its initial state, a very wide T_2 distribution, ranging from 0.1 ms to 900 ms, was observed, indicating a combination of large, medium, and small pores as well as some microporosity. This sample has a T_2 log-mean of around 27 ms.

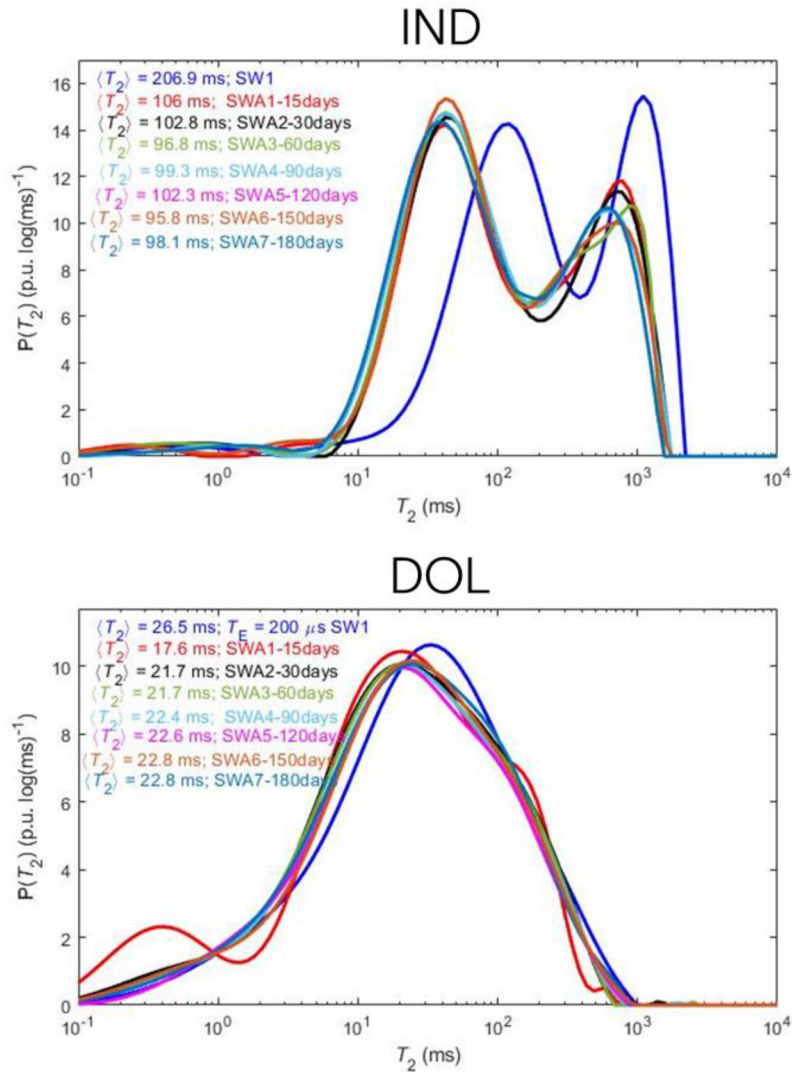


Figure 4—Effect of carbonate water flooding on NMR T2 distribution for IND and DOL samples

After each carbonated water flooding test, the NMR measurements were conducted to evaluate effects on T_2 distribution and therefore pore morphology (Ma, et al., 2021; Singer et al., 2023). The T_2 distributions for IND sample (Figure 4a) showed a clear shift towards shorter T_2 , indicating that the carbonated water had interacted with the pore surface and might have altered pore surface relaxivity, surface morphology and/or pore size distributions (Adenutsi et al., 2019; Singer et al., 2023; Al-Hamad et al., 2025a).

A clear shift in T_2 towards shorter time was also observed for the DOL samples. The shift, however, was less pronounced than for the IND sample. For DOL sample, the T_2 distribution was found to become wider, indicating possible increase in the varieties of small and large pores. This agrees with the increase in porosity observed for the DOL sample. A wider T_2 distribution might also indicate changes in pore connectivity, leading to a more heterogeneous pore system (Khather et al., 2022), altering permeability and resistivity.

Again, similar to permeability (Fig. 2) and resistivity (Fig. 3), the changes in NMR T_2 distributions, for both samples, was significant in the first 15-30 days of exposure, after which little noticeable shift was observed. Comparing the NMR results from this study with our previous study (Al-Hamad et al., 2025a), both static and dynamic experiments showed similar changes, i.e., a clear shift towards shorter T_2 .

Mechanical Properties

Rock mechanical properties were also assessed before and after carbonated water flooding. The samples were loaded in the acoustic cell and a net confining pressure of 1,500 psi was applied. The ultrasonic-based dynamic shear (G) and bulk (K) moduli were computed using the P- and S-waves velocities, and rock bulk density ρ ,

$$G = \rho V_s^2 \quad (3)$$

$$K = \rho \left(V_p^2 - \frac{4}{3} V_s^2 \right) \quad (4)$$

Results show that the DOL sample had higher initial acoustic velocities than the IND calcite sample (Table 3). After the carbonated water flooding, the calcite sample IND showed significant and continuous reductions in acoustic velocities (Table 3 and Figure 5), thus the dynamic mechanical properties (Eqs 3 and 4). For this rock sample, reductions of 15% in shear moduli and around 2% of the bulk moduli were observed. Such a reduction implies that the sample is becoming less rigid and more breakable (Lanzo et al., 1997). The DOL sample appears to maintain its mechanical integrity during the whole experiment, with <3% reduction in both shear and bulk moduli, despite the observed increase in the sample porosity and permeability. These observations suggest that the rock-carbonated water interactions are rock type dependent. Based on these limited results, it seems that the dolomitic formation is a better potential candidate for CO₂ storage than limestone formations.

Table 3—Measured Vp and Vs, and the calculated bulk and share moduli, before and after exposure with CO₂.

Sample	Period (Day)	Vp (m/s)	Vs (m/s)	G (GPa)	K (GPa)
IND	0 (before CO ₂)	4118	2130	10.84	26.05
	15	4050	2033	9.87	26.01
	30	4034	2029	9.84	25.75
	60	4017	2017	9.72	25.58
	90	3985	1992	9.48	25.28
	120	3989	1976	9.33	25.57
	150	3953	1968	9.25	25.77
	180	3937	1964	9.22	25.43
DOL	0 (before CO ₂)	4887	2560	16.26	37.57
	15	4838	2550	16.13	36.56
	30	4842	2553	16.17	36.60
	60	4842	2555	16.20	36.57
	90	4845	2554	16.18	36.66
	120	4840	2555	16.20	36.52
	150	4842	2550	16.13	36.65
	180	4839	2552	16.15	36.55

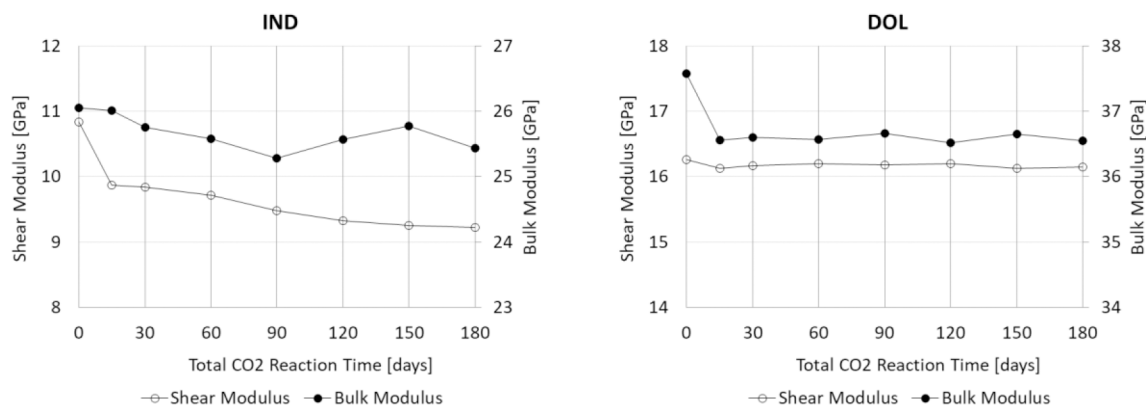


Figure 5—Effect of carbonated water on rock mechanical properties; shear and bulk modulus for IND and DOL samples.

It is noted that in our previous study (Al-Hamad et al., 2025a), under static ageing conditions, the rock mechanical stiffness for both rock samples (IND and DOL) were maintained with only minor observed decreases; reductions of <4% in shear and bulk moduli after 6 months of static ageing.

Rock Surface Imaging

High resolution images were acquired to capture the effect of carbonated water flooding on rock pore morphology. Images of the rock pore surfaces were taken using a DXR3 Raman Imaging microscope with a resolution of 0.5 microns and field of view of 1 mm x 0.8 mm imaging area. Images were captured after each carbonated water flooding stage. Figure 6 shows the images taken before and after the 6-month carbonated water flooding for the IND and DOL samples, respectively. Where chemical reactions occurred, the zone in which the rock appears in darker colors. The images clearly show some areas exhibiting dissolution and/or deformation after exposure to carbonated water, confirming and explaining some of the measurements and observations discussed earlier. For the DOL sample, the images taken after rock-carbonated water injection show, in some areas, enlargement in pore sizes, explaining the observed wider NMR T_2 distribution (Fig. 4). These observations are in line with what we observed in our previous study (Al-Hamad et al., 2025a) for IND and DOL samples, during static ageing.

The Fig. 6 image resolutions for the IND and DOL samples were 387×542 pixels and 387×509 pixels, respectively. Using the captured images, the changes in surface morphology were further analyzed and processed. Figure 7 displays the cumulative distribution of pixel intensities for images taken after six different times of exposure to carbonated water, including both histograms (in bars) and cumulative distribution functions (CDF, in lines) for the number of pixels across inverted intensity values, ranging from 0 (white) to 1 (black), the chemical reaction appears in darker colors.

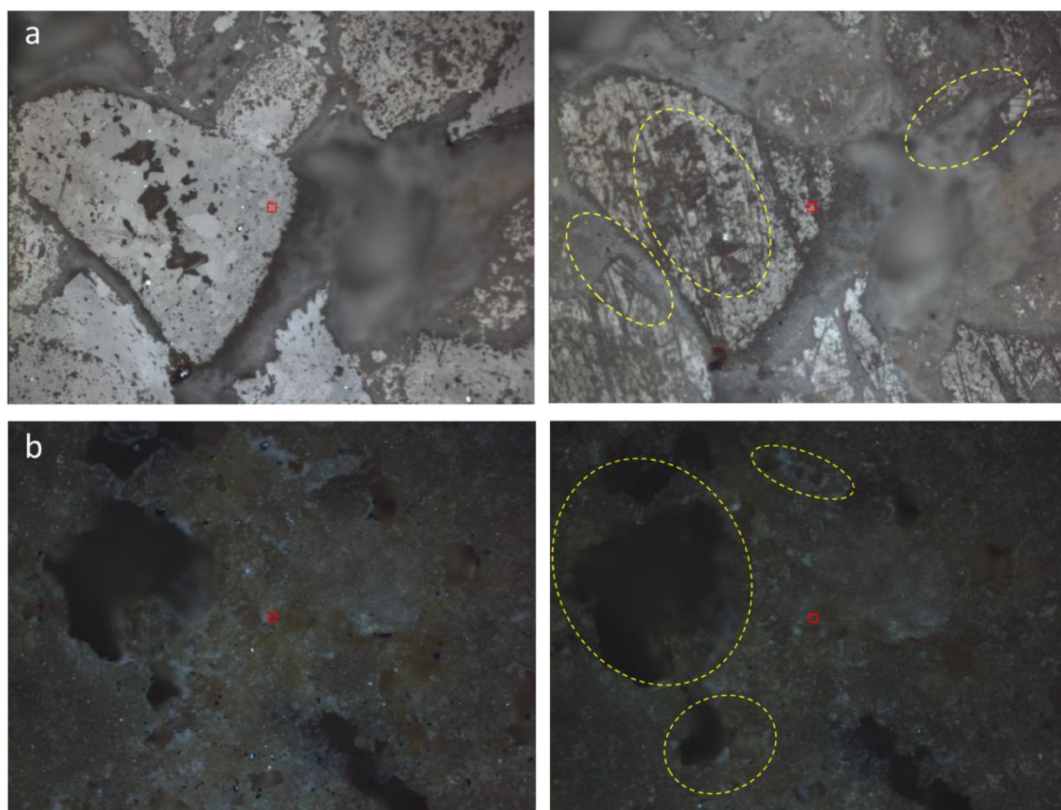


Figure 6—High resolution images before and after a 6-month period of CO₂ brine injection, for IND and DOL samples

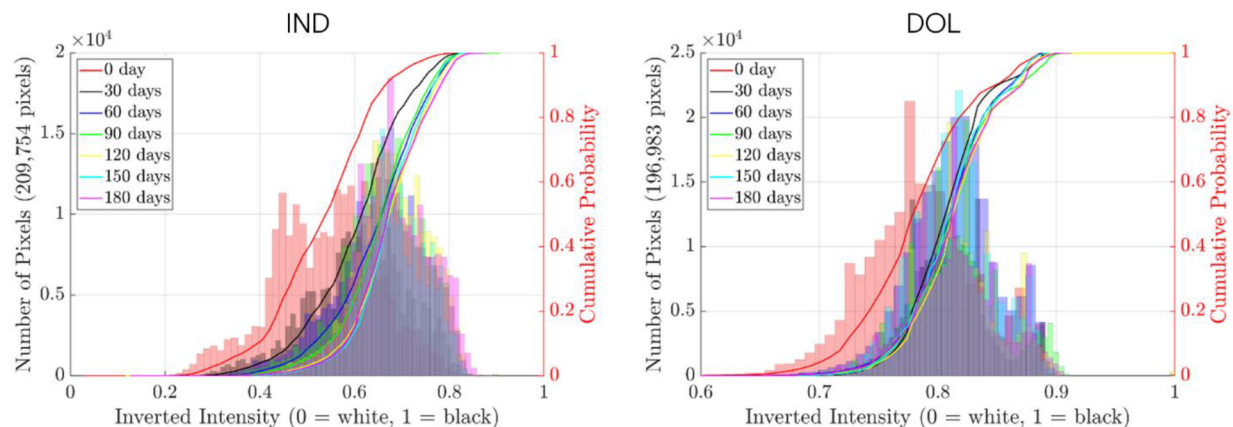


Figure 7—Carbonated water flooding shifting the distribution and cumulative distribution of pixel intensities extracted from the high resolution images of the IND and DOL sample to the right direction, or images become darker which indicating responses of rock pore surfaces to the acidic carbonate water.

Based on Fig. 6, even the grey scale may not fully capture subtle variations in the material. By comparing Fig. 7 histograms and CDFs across the time points, gradual shifts in the pixel intensity distribution can be observed, reflecting changes in the pores (dissolution/enlargement) over time. Similar to previous observations, the most significant changes occurred during early times of carbonated water exposure (around 30 days), then a slower rate of change was noticed after that time.

Water Analysis

Water chemistry was analyzed using a capillary electrophoresis unit with a UV/VIS detector and a cation program that the concentrations of the cations were calculated using established calibrated data. The measurement uncertainty was about $\pm 0.018\text{wt}\%$. In the middle of the long duration experiments, the water

analyzer failed, thus no additional water chemistry data were acquired after 4 months of carbonated water flooding.

Figure 8 shows the capillary electrophoresis chromatogram of the fresh carbonated brine, as well as the effluent brine sample after 15 days of carbonated water flooding. There are three peaks observed; the first one is for the sodium (Na^+) in the sample, the second one is for magnesium (Mg^{2+}) and the third is for calcium (Ca^{2+}). Figure 8 shows that the effluent sample collected from 15-days carbonated water flooding shows higher concentrations of Mg^{2+} and Ca^{2+} , indicating dissolution of rock matrices of the DOL sample.

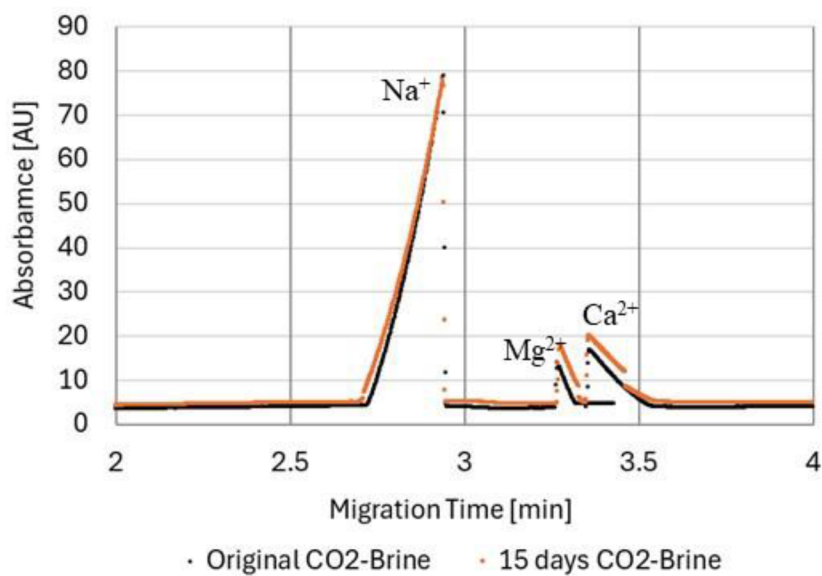


Figure 8—Water chemistry analyses of fresh carbonated brine, as well as brine effluent after 15 days of carbonated brine flooding through the DOL sample

Figures 9a and 9b show the changes of Ca^{2+} and Mg^{2+} concentration in the effluent brine vs. time, for both the IND and DOL samples. Results clearly show higher concentrations, despite fluctuations, of both Ca^{2+} and Mg^{2+} with carbonated water flooding, comparing to their starting point with fresh carbonated water.

It is interesting to note that since the IND is primarily calcite, changes in Mg^{2+} are minimal, while a higher Ca^{2+} concentration is continuously detected until the last stage of 120 days of carbonated water flooding, indicating a continuous dissolution of Ca^{2+} from the calcite mineral, with a maximum detected at 5%.

As for the DOL sample, the dissolution of Ca^{2+} rapidly occurred for 2.5 months, then it essentially stopped from happening. But the dissolution of Mg^{2+} continues until the end of the test at 120 days. The maximum increase in Mg^{2+} and Ca^{2+} concentrations was 18% and 6%, respectively.

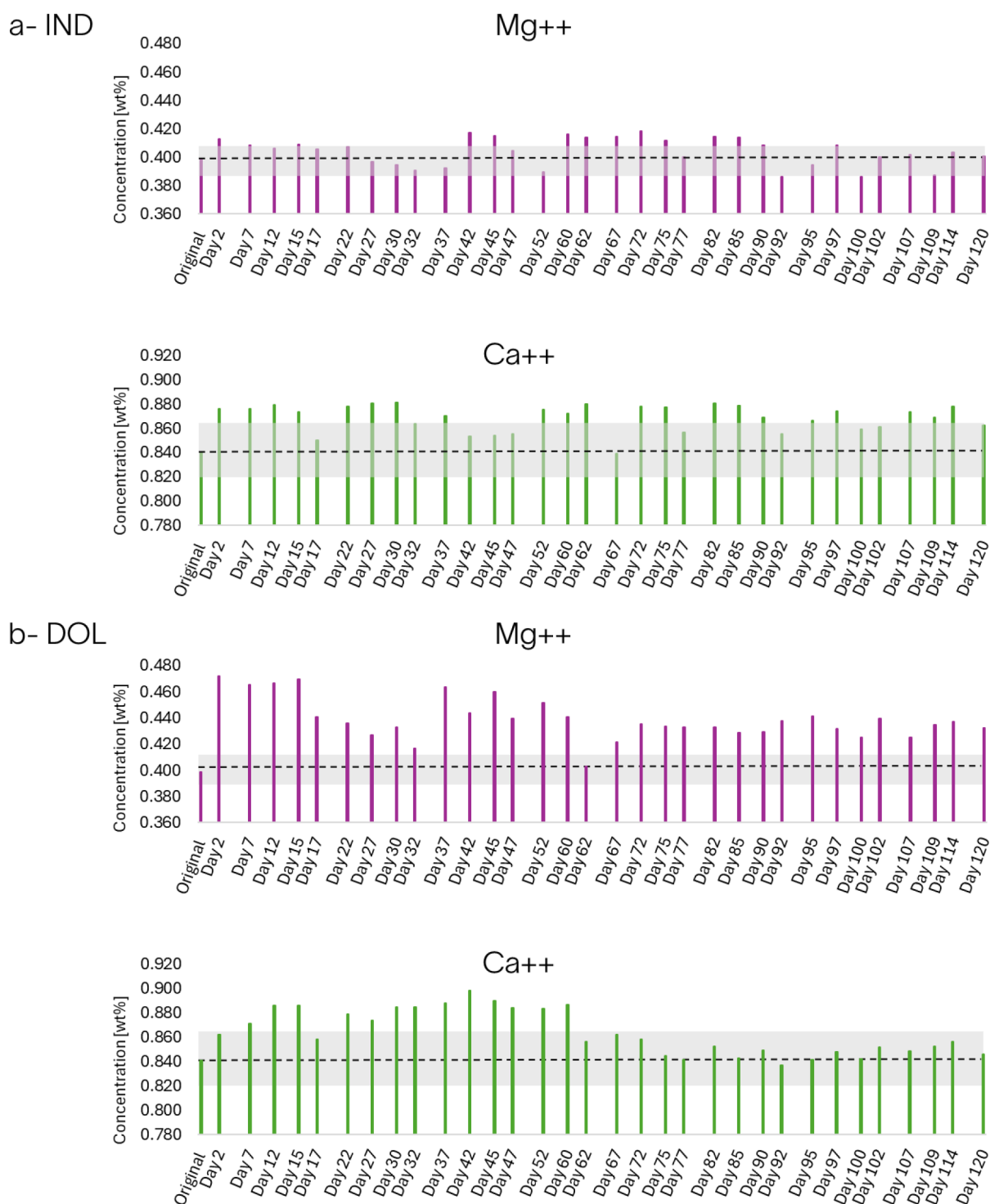


Figure 9—Water chemistry (Ca²⁺ and Mg²⁺) analyses of fresh carbonated brine, as well as brine effluent samples collected during the long duration carbonated water flooding, for IND and DOL samples. Note no data were acquired after 120 days due to a failure of water analyzer.

Conclusions

Two carbonate rock samples, one primarily calcite (IND) and another dolomite (DOL), were tested for effect of carbonated water dynamic ageing, up to 6 months, on rock properties, as compared to our earlier studies of static ageing (Singer et al., 2023; Al-Hamad et al., 2025). Experimental data of dynamic carbonated water ageing shows:

- Dynamic ageing has more profound effect than static ageing in rock permeability, resistivity, and rock mechanical properties.
- These more significant effects can also be observed from NMR measurements, as well as directly from high resolution images and image analyses.
- Water chemistry analyses of fresh carbonated water and effluent samples of carbonated water flooding experiments indicate continuous dissolution of Ca^{2+} in case of the calcite sample and Mg^{2+} with the dolomite sample. Ca^{2+} dissolution stopped for the dolomite sample after about 2.5 months of dynamic ageing.
- The differences or apparent contradictory, in permeability and resistivity changes with carbonates water interactions for the calcite IND and dolomite DOL samples (Figs. 2 and 3) demand a more systematic investigation.

Acknowledgement

The authors would like to thank Ahmed AlZoukani, Farhan Ali, and Azizah AlRadhwan for their efforts in executing laboratory measurements reported in this work, and Puvvala Vijaya for his support in preparing the cores and fluids used in this project.

References

- Adenutsi, C., Li, Z., Xu, Z., Sun, L., 2019. Influence of net confining stress on NMR T2 distribution and two-phase relative permeability, *Journal of Petroleum Science and Engineering*, **178**(766-777)
- Afkhami Karaei, M., Honarvar, B., Azdarpour, A., Mohammadian, E., 2020. CO2 Storage in Low Permeable Carbonate Reservoirs: Permeability and Interfacial Tension (IFT) Changes During CO2 Injection in an Iranian Carbonate Reservoir, *Periodica Polytechnica Chemical Engineering*, **64**(4), pp. 491–504
- Alam, M., Hjuler, M., Christensen, H., Fabricius, I., 2014. Petrophysical and rock-mechanics effects of CO2 injection for enhanced oil recovery: Experimental study on chalk from South Arne field, North Sea. *Journal of Petroleum Science and Engineering*, **122**, 468–487.
- Al-Hamad, M. Sindt, O., Ma, S., Abdallah, W., 2025a. Effect of CO2 Sequestration on Carbonate Formation Integrity, Paper SPWLA-2025-0131, *SPWLA 66th Annual Logging Symposium*, Dubai, UAE, May.
- Al-Hamad, et al, 2025b. Obtaining grain density of carbonate rocks: core plugs and powder samples, paper SPE-227052, MEOS GEO, 16-18 Sept., Bahrain.
- Ambrose, W., Lakshminarasimhan, S. 2008. Geologic factors controlling CO2 storage capacity and permanence: case studies based on experience with heterogeneity in oil and gas reservoirs applied to CO2 storage, *Environ Geol*, **54**:1619–1633
- Bacci, G., Korre, A., Durucan, S. 2011. Experimental investigation into salt precipitation during CO2 injection in saline aquifers, *Energy Procedia*, **4**:4450–4456.
- Bemer, E., Lombard, J. 2010. From injectivity to integrity studies of CO2 geological storage. *Oil & Gas Science and Technology - Rev. IFP*, **65**(3), 445–459.
- Burke, E. 2011. Understanding the Sensitivity of Different Drought Metrics to the Drivers of Drought under Increased Atmospheric CO2. *Journal of Hydrometeorology*, **12**, pp. 1378–1394.
- Endar, B.; Nurhandoko, B.; Prasetyo, A.; Hermawan, W.; Listyobudi, M.; Susilowati; Bajry, F.; Martha, R.K.; Supriyanto, E.B. Permeability, 2022. Change of Carbonate Rock Due to CO2 Injection for Simulating the Feasibility of CO2 Carbon Capture and Storage (CCS): A Case Study of Indonesian Carbonate. In *The 47th Annual Scientific Meeting of Himpunan Ahli Geofisika Indonesia*; HAGI.
- Gaus, I. 2010. Role and impact of CO2-rock interactions during CO2 storage in sedimentary rocks. *International Journal of Greenhouse Gas Control*, **4**(1): 73–89.
- Hamza, A., Hussein, I., Al-Marri, M., Mahmoud, M., Shawabkeh, R., Aparicio, S., 2021. CO2 enhanced gas recovery and sequestration in depleted gas reservoirs: A review, *Journal of Petroleum Science and Engineering* **196**, 107685.
- Holloway, S., and van der Straaten, R., 1995. The Joule II Project: The Underground Disposal of Carbon Dioxide, *Energy Convers. Manage.*, **36**, 6-9, 519–522
- Holm, L.W., 1959. Carbon Dioxide Solvent Flooding for Increased Oil Recovery. *Trans.* **216**: 225–231.
- Karvounis, P., Blunt, M., 2021. Assessment of CO2 geological storage capacity of saline aquifers under the North Sea. *International Journal of Greenhouse Gas Control* **111**, 103463

- Khather, M., Saeedi, A., Myers, M.B., Giwelli, A., Esteban, L., 2022. Impacts of Limestone Vertical Permeability Heterogeneity on Fluid-Rock Interaction During CCS. *Transp Porous Med* **142**, 435–451
- Kim, K., Vilarrasa, V., & Makhnenko, R.Y., 2018. CO₂ Injection Effect on Geomechanical and Flow Properties of Calcite-Rich Reservoirs. *Fluids*, **3**(3), 66.
- Kiseok, K., Vilarrasa, V., Makhnenko, R., 2018. CO₂ Injection Effect on Geomechanical and Flow Properties of Calcite-Rich Reservoirs, *Fluids* **3**, 3: 66.
- Lala, A., El-Sayed, N., 2017. Controls of pore throat radius distribution on permeability, *Journal of Petroleum Science and Engineering*, **157**(941-950).
- Lanzo, G., Vučetić, M., Doroudian, M., 1997. Reduction of Shear Modulus at Small Strains in Simple Shear. *Journal of Geotechnical and Geoenvironmental Engineering*, **123**(11).
- Li, H., Lau, H., Wei, X., Liu, S., 2021. CO₂ storage potential in major oil and gas reservoirs in the northern South China Sea. *International Journal of Greenhouse Gas Control* **108**, 103328.
- Ma, S. and Morrow, N.R., 1993. Similarity of pore structure and the selection of core samples for tests of scaling, *Proceedings of the 9th Wyoming EOR Symp*, May 12-13, Casper, Wyoming.
- Ma, S. and Morrow, N.R., 1996. Relationship between porosity and permeability for porous media, paper SCA-9610, *Int'l Symp of the Society of Core Analysts*, Montpellier, France, Sept 8-10.
- Ma, S., et al, 2006. Enhanced petrophysics: integration of core and log data for improved reservoir saturation monitoring, paper SPE-106350, *SPEKSA annual technical Symp*, May, Dhahran, Saudi Arabia.
- Ma, S., et al, 2013. Quality assurance of carbonate rock special core analysis- lesson learnt from a multi-year research project, Paper IPTC-16768, March 26-28, Beijing, China.
- Ma, S., et al, 2021. Objective-driven solid=surface-roughness characterization for enhanced nuclear-magnetic-resonance petrophysics, *SPE journal*, June.
- Major, J., Eichhubl, P., Dewers, T., Olson, J., 2018. Effect of CO₂-brine-rock interaction on fracture mechanical properties of CO₂ reservoirs and seals. *Earth and Planetary Science Letters*, **499**, pp. 37–47.
- Mohamed, I.M.M., and H.A.A. Nasr-El-Din, 2013. Fluid/Rock Interactions During CO₂ Sequestration in Deep Saline Carbonate Aquifers: Laboratory and Modeling Studies. *SPE J.* **18**.
- Ojala, I., 2011. The effect of CO₂ on the mechanical properties of reservoir and cap rock. *Energy Procedia*, **4**, pp. 5392–5397.
- Rosenbauer, R.J., Koksalan, T. 2022. Experimental determination of the solubility of CO₂ in electrolytes: application to CO₂ sequestration in deep, saline aquifers. *The Geological Society of America* **2**, p. 304.
- Rötting, T., Luquot, L., Carrera, J., Casalinuovo, D. 2015. Changes in porosity, permeability, water retention curve and reactive surface area during carbonate rock dissolution. *Chemical Geology*, **403**:86–98.
- Rukuni, T., Maree, J., Carlsson, F., 2015. Investigation of carbonate dissolution for the separation of magnesium hydroxide and calcium sulphate in a magnesium hydroxide-calcium sulphate mixed sludge, *African Journal Online*, **41**(2).
- Singer, G., et al, 2023. 2D surface roughness quantification for enhanced petrophysical applications, *SPE Journal*, Jan.
- Singer, G., et al, 2024. Laboratory study of CO₂ chemical reaction on the morphology of rock pore surfaces, Paper IPTC-23997, Feb 12-14, Dhahran, Saudi Arabia.
- Syed, S.A., et al, 2024. Relationship between rock acoustic dispersion and petrophysical properties for enhanced formation evaluation, paper IPTC-23747, Feb 12-14, Dhahran, Saudi Arabia
- Siqueira, T.A., Iglesias, R.S and Ketzer, J.M., 2017. Carbon dioxide injection in carbonate reservoirs - a review of CO₂-water-rock interaction studies. *Greenhouse Gases: Science and Technology*. **7**(5), 802–816 <https://doi.org/10.1002/ghg.1693>.
- Steel, L., Liu, Q., Mackay, E. & Maroto-Valer, M.M., 2016. CO₂ solubility measurements in brine under reservoir conditions: A comparison of experimental and geochemical modeling methods, *Greenhouse Gases: Science and Technology*, vol. **6**, no. 2, pp. 197–217.
- Tariq, Z., Abdulraheem, A., Elkatatny, S., Mahmoud, M., Muqtadir, A., Mobeen, M., 2018. Geomechanical Studies on CO₂ Sequestered Rocks in an Aqueous Saline Environment. Paper presented at the SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition, Dammam, Saudi Arabia
- Wang, H., Tian, L., Chai, X., Wang, J., Zhang, K., 2022 Effect of pore structure on recovery of CO₂ miscible flooding efficiency in low permeability reservoirs. *Journal of Petroleum Science and Engineering*, **208**, 109305.
- Wang, J. and Ehlig-Economides, C., 2023. Effect of predissolved natural gas on CO₂ solubility in water with various salinities at reservoir conditions, *Petrophysics journal*, **v64**(06).
- Zhang, Z., Agarwal, R., 2012. Numerical Simulation and Optimization of CO₂ Sequestration in Saline Aquifers. *The 6th International Conference on Energy Sustainability collocated with the ASME 2012 10th International Conference on Fuel Cell Science, Engineering and Technology*, pp. 795–803.